

Table 5: Detection results on the ICDAR 2015 dataset. The values in the bracket mean the height of the input images. “TB” and “PSE” are short for TextBoxes++ and PSENet.

Method	P	R	F	FPS
CTPN (?)	74.2	51.6	60.9	7.1
EAST (?)	83.6	73.5	78.2	13.2
SSTD (?)	80.2	73.9	76.9	7.7
WordSup (?)	79.3	77	78.2	-
Corner (?)	94.1	70.7	80.7	3.6
TB (?)	87.2	76.7	81.7	11.6
RRD (?)	85.6	79	82.2	6.5
MCN (?)	72	80	76	-
TextSnake (?)	84.9	80.4	82.6	1.1
PSE-1s (?)	86.9	84.5	85.7	1.6
SPCNet (?)	88.7	85.8	87.2	-
LOMO (?)	91.3	83.5	87.2	-
CRAFT (?)	89.8	84.3	86.9	-
SAE(720) (?)	85.1	84.5	84.8	3
SAE(990) (?)	88.3	85.0	86.6	-
DB-ResNet-18 (736)	86.8	78.4	82.3	48
DB-ResNet-50 (736)	88.2	82.7	85.4	26
DB-ResNet-50 (1152)	91.8	83.2	87.3	12

50” surpasses the previous state-of-the-art method by 1.9% and 3.8% on the MSRA-TD500 and MLT-2017 dataset respectively. For the speed, “DB-ResNet-50” is 3.2 times faster than the previous fastest method (?) on the MSRA-TD500 dataset. With a light-weight backbone, “DB-ResNet-18 (736)” achieves a comparative accuracy compared to the previous state-of-the-art method (?) (82.8 vs 83.0) and runs at 62 FPS, which is 6.2 times faster than the previous fastest method (?), on the MSRA-TD500. The speed can be further accelerated to 82 FPS (“ResNet-18 (512)”) by decreasing the input size.

Limitation

One limitation of our method is that it can not deal with cases “text inside text”, which means that a text instance is inside another text instance. Although the shrunk text regions are helpful to the cases that the text instance is not in the center region of another text instance, it fails when the text instance exactly locates in the center region of another text instance. This is a common limitation for segmentation-based scene text detectors.

Conclusion

In this paper, we have presented a novel framework for detecting arbitrary-shape scene text, which includes the proposed differentiable binarization process (DB) in a segmentation network. The experiments have verified that our method (ResNet-50 backbone) consistently outperforms the state-the-art methods on five standard scene text benchmarks, in terms of speed and accuracy. In particular, even with a lightweight backbone (ResNet-18), our method can achieve competitive performance on all the testing datasets

Table 6: Detection results on the MSRA-TD500 dataset. The values in the bracket mean the height of the input images.

Method	P	R	F	FPS
(?)	71	61	69	-
DeepReg (?)	77	70	74	1.1
RRPN (?)	82	68	74	-
RRD (?)	87	73	79	10
MCN (?)	88	79	83	-
PixelLink (?)	83	73.2	77.8	3
Corner (?)	87.6	76.2	81.5	5.7
TextSnake (?)	83.2	73.9	78.3	1.1
(?)	83.0	77.4	80.1	-
(?)	87.4	76.7	81.7	-
CRAFT (?)	88.2	78.2	82.9	8.6
SAE (?)	84.2	81.7	82.9	-
DB-ResNet-18 (512)	85.7	73.2	79.0	82
DB-ResNet-18 (736)	90.4	76.3	82.8	62
DB-ResNet-50 (736)	91.5	79.2	84.9	32

Table 7: Detection results on the MLT-2017 dataset. Methods with “*” are collected from (?). The images in the MLT-2017 dataset are re-sized to 768×1024 in our method. “PSE” is short for PSENet.

Method	P	R	F	FPS
SARL_FDU_RRP_N_V1*	71.2	55.5	62.4	-
Sensetime_OCR*	56.9	69.4	62.6	-
SCUT_DLVLab1*	80.3	54.5	65.0	-
e2e_ctc01_multi_scale*	79.8	61.2	69.3	-
Corner (?)	83.8	55.6	66.8	-
PSE (?)	73.8	68.2	70.9	-
DB-ResNet-18	81.9	63.8	71.7	41
DB-ResNet-50	83.1	67.9	74.7	19

with real-time inference speed. In the future, we are interested in extending our method for end-to-end text spotting.

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